

ENGR 102 Summer Bridge

Day 1 Instructor Key

Exact Output, Stored Values, and First Complete Program

20 points • approximately 20-25 minutes

Name		UIN		Section	
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Boolean-operator convention. Python operators appear as [and](#), [or](#), and [not](#). Their lowercase spelling is required in code.

Show intermediate state for trace credit. Program credit requires one coherent solution. Unless a prompt says otherwise, give exact Python 3 output.

Question 1. Trace stored state and exact output.

4 points

```
label = "VIP"
count = 2
print(label)
count = count + 3
print("count:", count)
label = "Bridge"
print(label, count)
```

Show the stored values after each assignment, then write every output line exactly.

Answer and explanation

label begins as VIP and count begins as 2. The update stores 5 in count; the final assignment replaces VIP with Bridge. Exact output lines are VIP; count: 5; Bridge 5.

Question 2. Separate a copied value from a later overwrite.

4 points

```
stored = "ready"
print("first", stored)
displayed = stored
stored = "review"
print("second", displayed, stored)
```

Trace stored and displayed. Explain why displayed does not change when stored is overwritten.

Answer and explanation

displayed receives the current text ready before stored is overwritten. Reassigning stored does not retroactively change displayed. Exact output is first ready, then second ready review.

Question 3. Repair one exact-format defect.

4 points

```
course = "ENGR 102"
meeting = "Bridge"
print("Course", course)
print("Meeting:", meeting)
```

The first line must be Course: ENGR 102. Give the actual output, the minimal repair, and the corrected exact output.

Answer and explanation

The actual first line is Course ENGR 102 because the label has no colon. Replace the first print with print("Course:", course). The corrected lines are Course: ENGR 102 and Meeting: Bridge.

Question 4. Program construction**8 points**

Write one complete Python program that satisfies every requirement.

- Store Maya in name, 4 in kits, and 6 in items_per_kit.
- Compute total_items from the two numeric variables.
- Print exactly Student: Maya, Kits: 4, and Total items: 24 on three separate lines.
- Use the variables in the print statements; do not type 24 as the result.

Answer and explanation

```
name = "Maya"
kits = 4
items_per_kit = 6
total_items = kits * items_per_kit

print("Student:", name)
print("Kits:", kits)
print("Total items:", total_items)
```

The calculation is stored before output so the visible total is evidence from `kits * items_per_kit`, not a copied answer. Exact output has three lines: Student: Maya; Kits: 4; Total items: 24. Python inserts one space between comma-separated print arguments.

Scoring guidance

2 named values, 2 calculated total, 3 exact labeled lines in order, 1 uses variables rather than a copied result.